Mir Mahathir Mohammad - Curriculum Vitae

Contact

Location: Salt Lake City, UT **Email:** mirmahathir1@gmail.com **Portfolio:** https://mirmahathir1.github.io/

LinkedIn GitHub Google Scholar DBLP

Education

University of Utah, Salt Lake City, Utah
Ph.D. in Computer Science
Kahlert School of Computing | August 2023 – Present
Advisor: El Kindi Rezig

University of Utah, Salt Lake City, Utah
M.S. in Computer Science
Kahlert School of Computing | August 2023 – December 2025
CGPA: 3.978 / 4.0
Advisor: El Kindi Rezig

Bangladesh University of Engineering and Technology (BUET), Dhaka, Bangladesh
B.Sc. in Computer Science and Engineering
April 2017 – April 2022
Advisor: Muhammad Abdullah Adnan

Research Interests

I focus on data preparation in general and data discovery in particular for data lakes. I built SemDisc, the first end-to-end semantic join discovery system using a query-by-example interface. I also explored how discovered relationships can recommend meaningful ways to organize and sort data. I built NLDisc, which turns a user's natural language description of intent into join discovery over data lakes, and co-developed LORY-J, which treats spatial attributes as first-class signals to discover location-aware joins. I contributed to Buckaroo, a visual data wrangling system that enables users to interactively clean and repair data anomalies in visualizations. My interests include:

Data Systems • Data Lakes • Data Discovery & Integration • Data Wrangling • Data Cleaning • AI for Data Management

Publications

Publications since starting Ph.D. (2023-Present, ~3.2 years)

- ***SIGMOD'26***: Mir Mahathir Mohammad, El Kindi Rezig. "Qualitative Join Discovery in Data Lakes using Examples." Accepted at ACM SIGMOD 2026. [PDF] [Link] *a system for discovering hybrid join paths (combining semantic and equi-joins) in data lakes using query-by-example, supporting hidden tables and semantic tuple matching*
- ***SIGMOD'26***: Mir Mahathir Mohammad, El Kindi Rezig. "SemDisc: An End-to-End Query-by-Example Semantic Join Discovery System." Accepted at ACM SIGMOD 2026. [PDF] [Link] *an interactive system enabling users to discover hybrid join paths in schema-less data lakes through a query-by-example interface*
- ***SIGMOD'26***: Mir Mahathir Mohammad, El Kindi Rezig. "NLDisc: Intent-aware Join Discovery using Natural Language." Accepted at ACM SIGMOD 2026. [PDF] [Link] *a system for intent-aware join discovery over data lakes using natural language queries with interactive intent disambiguation and join path diversification*
- ***VLDB'26***: Lu Xing, Mir Mahathir Mohammad, Walid G. Aref, El Kindi Rezig. "LORY-J: Location-aware Join Discovery from Data Lakes." Accepted at VLDB 2026. [PDF] [Link] *a system for location-aware join discovery in data lakes that treats spatial attributes as first-class signals, composing hybrid join paths through an interactive, map-driven interface*
- ***CIDR'26***: El Kindi Rezig, Mir Mahathir Mohammad, Nicolas Baret, Ricardo Mayerhofer, Andrew McNutt, Paul Rosen. "Towards Scalable Visual Data Wrangling via Direct Manipulation." Accepted at CIDR 2026. [PDF] [Link] *a visual data wrangling system that enables users to clean and repair data anomalies through direct manipulation of interactive visualizations*
- ***VLDB'25***: Akash Khatri, Mir Mahathir Mohammad, El Kindi Rezig. "Sort it Like You Mean It: Discovering Semantically Interesting Attribute Augmentations to Sort Tables." Accepted at VLDB 2025. [PDF] [Link] *Recommends semantically meaningful ways to sort tables by automatically discovering and augmenting attributes from data lakes using LLMs.*

Undergraduate research

- ***IEEE FG'24***: Iftekhar E Mahbub Zeeon, Mir Mahathir Mohammad, Muhammad Abdullah Adnan. "BTVSL: A Novel Sentence-Level Annotated Dataset for Bangla Sign Language Translation." Accepted at IEEE FG 2024. *Introduces the first large-scale sentence-level dataset for Bangla Sign Language translation.* [PDF] [Link]
- ***Neurocomputing'22***: Md. Ashraful Islam, Mir Mahathir Mohammad, Sarkar Snigdha Sarathi Das, Mohammed Eunus Ali. "A survey on deep learning based Point-of-Interest (POI) recommendations." Accepted at Neurocomputing (Journal), 2022. [PDF] [Link] *Categorizes deep learning approaches for POI recommendation systems in location-based social networks.*

Research Experience

University of Utah, Kahlert School of Computing, Salt Lake City, UT

Graduate Research Assistant, August 2023 – Present

Advisor: El Kindi Rezig (Postdoc, MIT CSAIL; Ph.D., Purdue University)

- Developed algorithms for qualitative join discovery in data lakes using example-based queries
- Built systems for semantic table sorting, natural-language intent-aware join discovery, and location-aware join discovery over data lakes
- Implemented scalable data wrangling systems with direct manipulation interfaces, handling data transformations

Bangladesh University of Engineering and Technology, CSE, Dhaka, Bangladesh

Research Assistant, July 2022 – June 2023

Advisor: Muhammad Abdullah Adnan (Ph.D., University of California San Diego)

- Developed machine learning pipelines for processing and analyzing video datasets for sign language translation
- Built data collection and annotation systems for creating structured datasets, handling data cleaning

Technical Skills

LLMs/ML: PyTorch, LangChain, Finetuning Language Models (Flan-T5-base, Qwen2.5-0.5B, Llama3.2-1B)

DevOps: Docker, Kubernetes

SWE: Full Stack Web Development (Vue.js, React.js, Next.js, Express.js), Streamlit, Progressive Web App (PWA) Development for Android

Additional Experience

Everforth, Tokyo, Japan (Remote)

Frontend Developer, April 2022 – June 2023

- Built e-commerce web applications using Vue.js and CakePHP